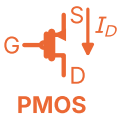


NMOS



PMOS

lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

ANALOG LAB · LECTURES 23 – 25

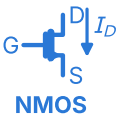
The CMOS Inverter from VTC to delay

Every derivation from your notes worked line by line, the closed forms worth memorising, and questions for you to solve before you read the answers.

Part 0	Prerequisites: square law with signs, the critical voltages, the maths you need
Part 1 · L23	The VTC: regions A–E, and deriving V_{IL} and V_{IH}
Part 2 · L24	V_{th}, k_R, sizing, the symmetric inverter, noise margins, layout
Part 3 · L25	Load capacitance, delay definitions, average-current and state-equation methods
Part 4	Formula sheet, memory tricks, flashcards, exam playbook

HOW TO USE THIS GUIDE

Orange **Your turn** boxes are questions: cover the page below them and solve first. Full solutions are at the end of each part. Blue boxes hold the results to memorise. The strip at the top of every page carries the MOSFET equations, so you never need to flip back for them.



lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

PART 0

Prerequisites

Everything Lectures 23–25 assume you already know

0.1 The square law, with signs

Kang & Leblebici notation, which your notes use: $V_{T0,n} > 0$, $V_{T0,p} < 0$, and the strength of a transistor is $k = \mu C_{ox}\, W/L$. The PMOS uses **the same formulas as the NMOS**. Its voltages are just all negative.

REGION	NMOS (ALL POSITIVE)	PMOS (ALL NEGATIVE)
Cut-off	$V_{GS} < V_{T0,n}$, $I_D = 0$	$V_{GS} > V_{T0,p}$, $I_D = 0$
Linear	$V_{DS} < V_{GS} - V_{T0,n}$	$V_{DS} > V_{GS} - V_{T0,p}$
	$I_D = \frac{k}{2}\left[2(V_{GS} - V_{T0})V_{DS} - V_{DS}^2\right]$	
Saturation	$V_{DS} \geq V_{GS} - V_{T0,n}$	$V_{DS} \leq V_{GS} - V_{T0,p}$
	$I_D = \frac{k}{2}(V_{GS} - V_{T0})^2$	

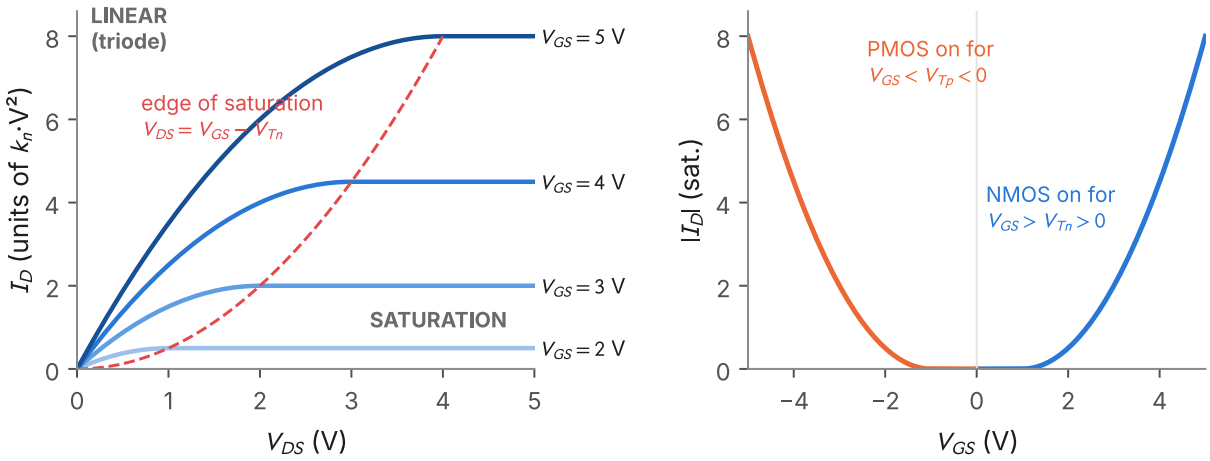
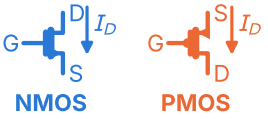


Fig. 0.1 Left: NMOS I_D–V_{DS} curves. The dashed red parabola $V_{DS} = V_{GS} - V_{T0,n}$ separates linear (left) from saturation (flat, right). Right: the PMOS is the mirror image, and turns on for negative V_{GS}.

THE PMOS INEQUALITY FLIP

For the PMOS, every "$\geq$" of the NMOS becomes "$\leq$", because the voltages are negative. If that feels slippery, work in magnitudes: PMOS is saturated when $|V_{DS}| \geq |V_{GS}| - |V_{T0,p}|$, exactly like the NMOS.



lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

0.2 Critical voltages and noise margins

The **voltage transfer characteristic** (VTC) plots V_{out} against V_{in}. Five numbers describe it:

- V_{OH} and V_{OL}: the output high and low levels.
- V_{IL} and V_{IH}: the two inputs where the slope is $dV_{out}/dV_{in} = -1$.
- V_{th}: the switching threshold, where $V_{out} = V_{in}$.

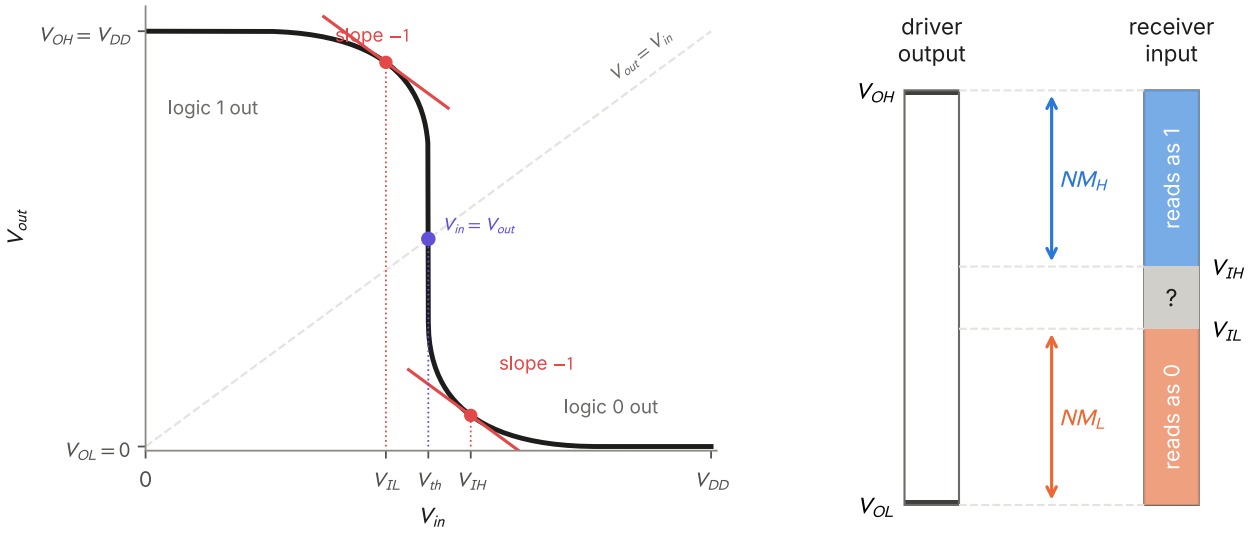


Fig. 0.2 Left: a VTC with its five critical voltages. Right: noise margins. A driver outputs at least V_{OH} for a 1. The receiver reads anything above V_{IH} as 1. The gap between them is how much noise a 1 can tolerate.

MEMORISE

$$NM_L = V_{IL} - V_{OL} \qquad NM_H = V_{OH} - V_{IH}$$

WHY THE SLOPE IS −1 AT THE EDGES

A small noise Δ on the input appears at the output as $|dV_{out}/dV_{in}| \cdot \Delta$. Where $|\text{slope}| < 1$, the noise shrinks. Where $|\text{slope}| > 1$, it grows. V_{IL} and V_{IH} mark where one regime turns into the other.

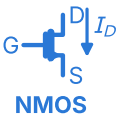
0.3 The two pieces of maths you need

(a) Implicit differentiation (Lecture 23)

In the VTC, V_{out} is a function of V_{in}. So when you differentiate a current equation with respect to V_{in}, every V_{out} term picks up a factor dV_{out}/dV_{in} (the chain rule):

$$\frac{d}{dV_{in}}(V_{out} - V_{DD})^2 = 2(V_{out} - V_{DD})\frac{dV_{out}}{dV_{in}}, \qquad \frac{d}{dV_{in}}\Big[(V_{in} - a)V_{out}\Big] = V_{out} + (V_{in} - a)\frac{dV_{out}}{dV_{in}}$$

Then you substitute $dV_{out}/dV_{in} = -1$.



lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

(b) Two integrals (Lecture 25)

$$\int dV = V \qquad \text{and} \qquad \int \frac{dV}{V(a-V)} = \frac{1}{a}\ln\frac{V}{a-V} \qquad \left(\text{since } \frac{1}{V(a-V)} = \frac{1}{a}\left[\frac{1}{V} + \frac{1}{a-V}\right]\right)$$

YOUR TURN · PART 0 (SOLUTIONS JUST BELOW)

0.1 An NMOS has $V_{T0,n} = 0.6$ V and $k_n = 200\,\mu$A/V². It is biased at $V_{GS} = 1.5$ V and $V_{DS} = 0.5$ V. Which region is it in, and what is I_D?

0.2 In an inverter with $V_{DD} = 3.3$ V, the PMOS has $V_{T0,p} = -0.7$ V and $k_p = 80\,\mu$A/V². Take $V_{in} = 1.0$ V and $V_{out} = 2.9$ V. Find $V_{GS,p}$ and $V_{DS,p}$, the region, and $|I_D|$.

0.3 Differentiate $2(V_{in} - V_{DD} - V_{T0,p})(V_{out} - V_{DD})$ with respect to V_{in}.

0.4 Take $V_{OH} = 3.3$, $V_{OL} = 0.2$, $V_{IL} = 1.2$ and $V_{IH} = 1.9$ V. Find both noise margins. Which logic level is better protected?

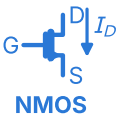
Solutions · Part 0

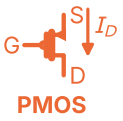
0.1 Its overdrive is $V_{GS} - V_{T0,n} = 0.9$ V. Since $V_{DS} = 0.5 < 0.9$, it is in the **linear** region: $I_D = 100\,\mu\,[2(0.9)(0.5) - 0.25] = 100\,\mu \times 0.65 = \mathbf{65\ \mu A}$.

0.2 $V_{GS,p} = 1.0 - 3.3 = -2.3$ V and $V_{DS,p} = 2.9 - 3.3 = -0.4$ V, so $V_{GS,p} - V_{T0,p} = -1.6$ V. Since $-0.4 > -1.6$, it is **linear**: $I_D = 40\,\mu\,[2(-1.6)(-0.4) - 0.16] = 40\,\mu \times 1.12 = \mathbf{44.8\ \mu A}$.

0.3 Use the product rule: $2\left[(V_{in} - V_{DD} - V_{T0,p})\frac{dV_{out}}{dV_{in}} + (V_{out} - V_{DD})\right]$. This is the middle term of the lecture's step after eq. ①.

0.4 $NM_L = 1.2 - 0.2 = \mathbf{1.0\ V}$ and $NM_H = 3.3 - 1.9 = \mathbf{1.4\ V}$. The logic 1 is better protected.





lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

PART 1 · LECTURE 23

The CMOS inverter VTC

Two complementary switches, five regions, and the slope −1 derivations

1.1 The circuit

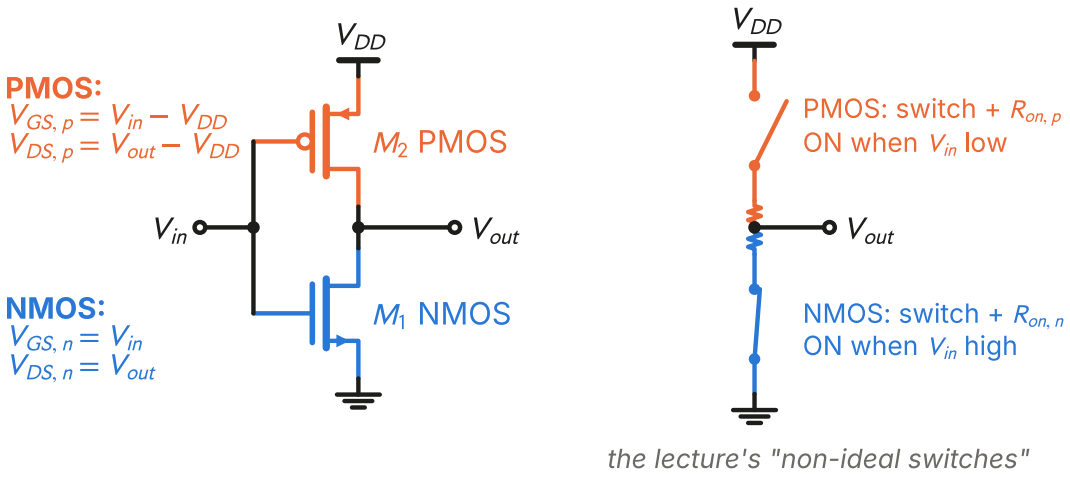


Fig. 1.1 The CMOS inverter: the NMOS M_1 pulls down and the PMOS M_2 pulls up. Both gates share V_{in}, and both drains form V_{out}. Right: the lecture's model, two non-ideal switches (each a switch plus its on-resistance) that are never both on in steady state.

Reading the terminal voltages off the figure is the first step of every CMOS question. Each one translates a transistor condition into a condition on V_{in} and V_{out}:

$$V_{GS,n} = V_{in}, \quad V_{DS,n} = V_{out}, \qquad V_{GS,p} = V_{in} - V_{DD} = -(V_{DD} - V_{in}), \quad V_{DS,p} = V_{out} - V_{DD} = -(V_{DD} - V_{out})$$

1.2 The two saturation boundaries

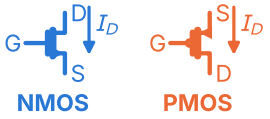
1. **NMOS saturated** needs $V_{GS,n} \geq V_{T0,n}$, i.e. $V_{in} \geq V_{T0,n}$. It also needs $V_{DS,n} \geq V_{GS,n} - V_{T0,n}$:

$$\boxed{V_{out} \geq V_{in} - V_{T0,n}} \quad \text{(NMOS sat. above this line)}$$

2. **PMOS saturated** needs $V_{GS,p} \leq V_{T0,p}$, i.e. $V_{in} - V_{DD} \leq V_{T0,p}$, i.e. $V_{in} \leq V_{DD} + V_{T0,p}$. It also needs $V_{DS,p} \leq V_{GS,p} - V_{T0,p}$:

$$V_{out} - \cancel{V_{DD}} \leq V_{in} - \cancel{V_{DD}} - V_{T0,p} \;\Rightarrow\; \boxed{V_{out} \leq V_{in} - V_{T0,p}} \quad \text{(PMOS sat. below this line)}$$

Both boundaries are straight lines of slope 1. The NMOS line crosses the axis at $V_{in} = V_{T0,n}$. The PMOS line sits $|V_{T0,p}|$ higher. Draw them on the (V_{in}, V_{out}) plane and the VTC passes through every region:



lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

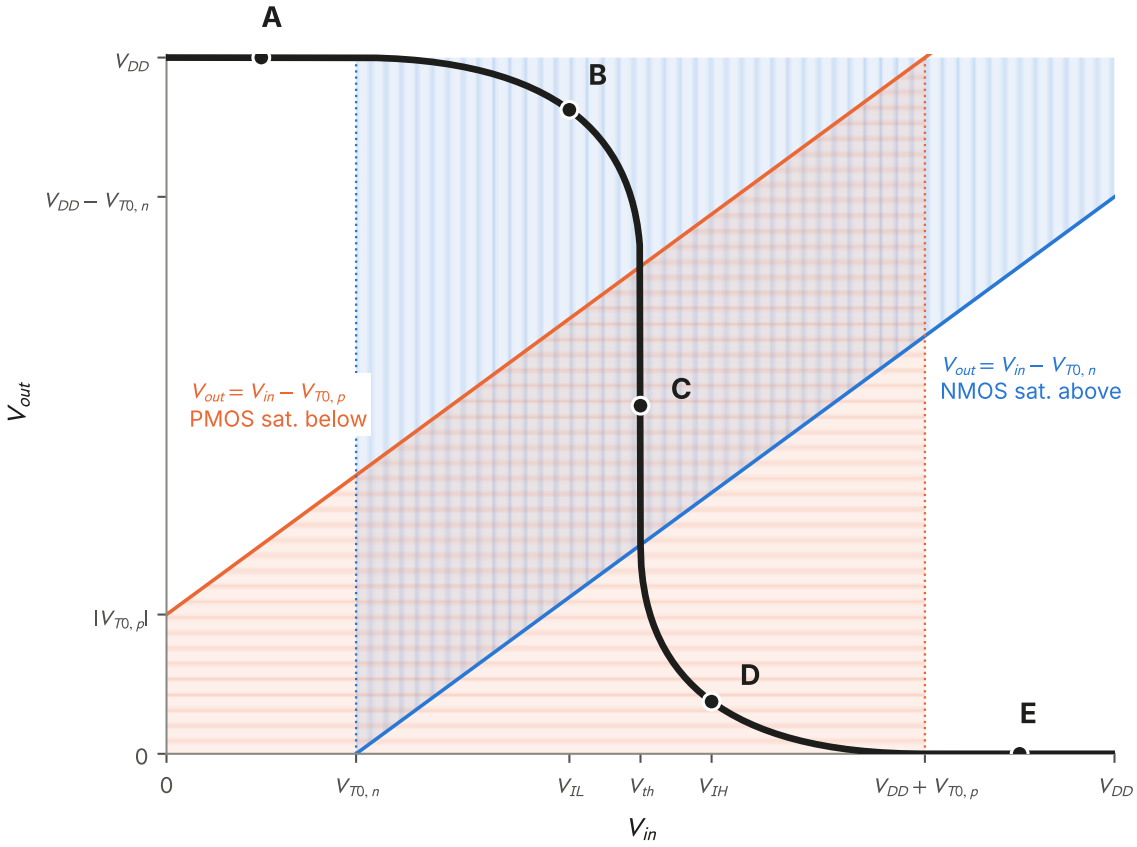
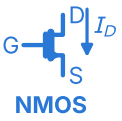
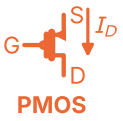


Fig. 1.2 The lecture's hatched map, computed exactly ($V_{DD} = 5$ V, $V_{T0,n} = 1$ V, $V_{T0,p} = -1$ V, $k_R = 1$). Blue vertical hatching: NMOS saturated. Orange horizontal hatching: PMOS saturated. Where they cross, both are saturated, and the VTC drops vertically through C.





lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

1.3 The five regions

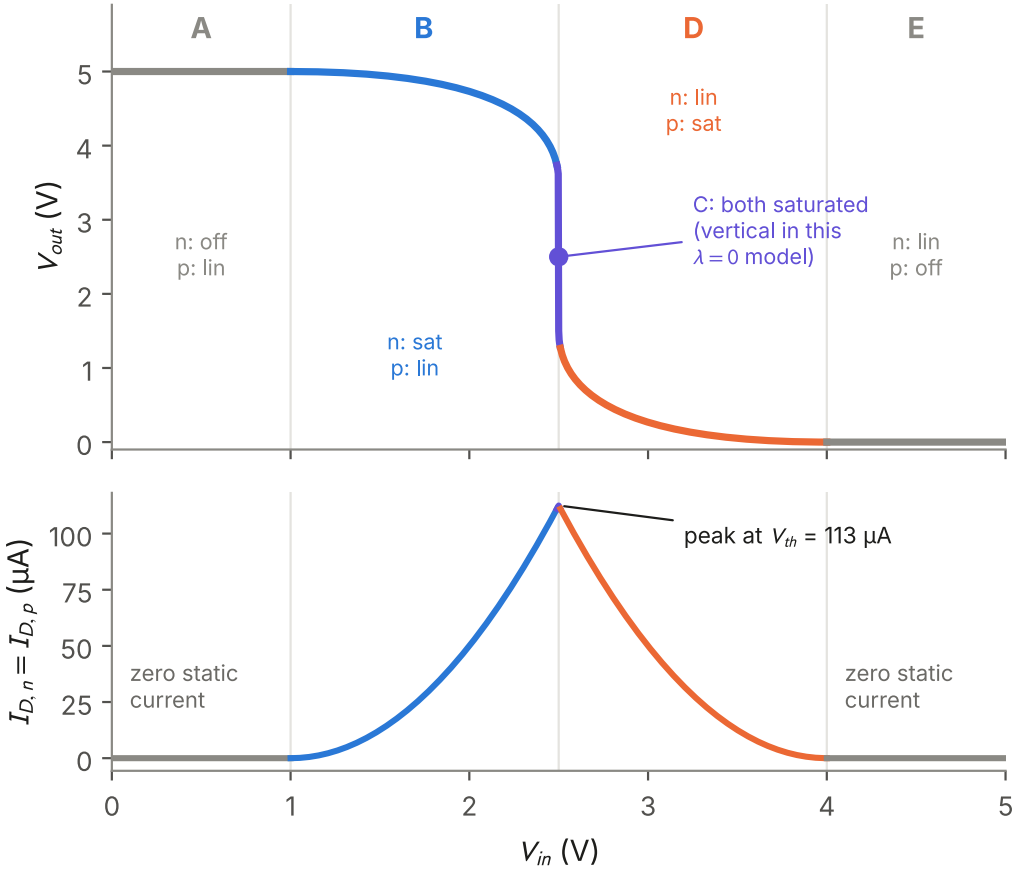
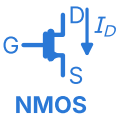


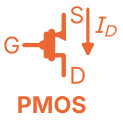
Fig. 1.3 Top: the VTC coloured by region. Bottom: the current through both transistors. It is zero at both ends, so a CMOS inverter draws no static power, and it peaks at V_{th}.

POINT	INPUT RANGE	NMOS	PMOS	WHAT YOU GET
A	$V_{in} < V_{T0,n}$	cut-off	linear	$V_{OH} = V_{DD}$ (no current, so no drop across the PMOS)
B	$V_{T0,n} \leq V_{in} < V_{th}$	saturation	linear	V_{IL} lies here
C	$V_{in} = V_{th}$	saturation	saturation	V_{th} (Lecture 24)
D	$V_{th} < V_{in} \leq V_{DD} + V_{T0,p}$	linear	saturation	V_{IH} lies here
E	$V_{in} > V_{DD} + V_{T0,p}$	linear	cut-off	$V_{OL} = 0$

MEMORY TRICK

Walk from left to right: the transistor that is **turning on** is saturated, and the one that is **holding the output** near its rail is linear. In B, the NMOS is just waking up (saturated) while the PMOS still holds V_{out} near V_{DD} (linear). D is the mirror image.





lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

1.4 Deriving V_{IL} (point B: NMOS sat., PMOS linear)

1. **Equate the currents**, $I_{D,n} = I_{D,p}$. Use the saturation formula for the NMOS and the linear formula for the PMOS, then divide by $k_p/2$ with $k_R = k_n/k_p$:

$$k_R(V_{in} - V_{T0,n})^2 = 2\underbrace{(V_{in} - V_{DD} - V_{T0,p})}_{V_{GS,p} - V_{T0,p}}\underbrace{(V_{out} - V_{DD})}_{V_{DS,p}} - (V_{out} - V_{DD})^2 \qquad (1)$$

2. **Differentiate with respect to** V_{in} (the chain rule from §0.3) and cancel the common 2:

$$k_R(V_{in} - V_{T0,n}) = (V_{in} - V_{DD} - V_{T0,p})\frac{dV_{out}}{dV_{in}} + (V_{out} - V_{DD}) - (V_{out} - V_{DD})\frac{dV_{out}}{dV_{in}}$$

3. **Set** $V_{in} = V_{IL}$ **and** $dV_{out}/dV_{in} = -1$**:**

$$k_R(V_{IL} - V_{T0,n}) = -V_{IL} + V_{DD} + V_{T0,p} + V_{out} - V_{DD} + V_{out} - V_{DD}$$

4. **Collect the** V_{IL} **terms** on the left, $(1 + k_R)V_{IL}$, and divide:

$$\boxed{V_{IL} = \frac{2V_{out} + V_{T0,p} - V_{DD} + k_R V_{T0,n}}{1 + k_R} \qquad (2)}$$

5. **Solve (1) and (2) together.** Rearrange (2) for V_{out}, substitute it into (1), and you get a quadratic in V_{IL}. Keep the root that lies between $V_{T0,n}$ and V_{th}.

$$V_{out}\big|_{V_{IL}} = \tfrac{1}{2}\left[(1 + k_R)V_{IL} + V_{DD} - V_{T0,p} - k_R V_{T0,n}\right]$$

1.5 Deriving V_{IH} (point D: NMOS linear, PMOS sat.)

1. **Equate the currents** with the roles swapped:

$$k_R\left[2(V_{in} - V_{T0,n})V_{out} - V_{out}^2\right] = (V_{in} - V_{DD} - V_{T0,p})^2 \qquad (3)$$

2. **Differentiate** and cancel the 2s:

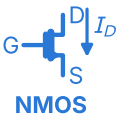
$$k_R\left[(V_{in} - V_{T0,n})\frac{dV_{out}}{dV_{in}} + V_{out} - V_{out}\frac{dV_{out}}{dV_{in}}\right] = V_{in} - V_{DD} - V_{T0,p}$$

3. **Set** $V_{in} = V_{IH}$ **and slope** $= -1$**:** $k_R\left[-V_{IH} + V_{T0,n} + 2V_{out}\right] = V_{IH} - V_{DD} - V_{T0,p}$

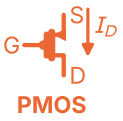
4. **Collect the terms:** $k_R(2V_{out} + V_{T0,n}) + V_{DD} + V_{T0,p} = V_{IH}(1 + k_R)$, so

$$\boxed{V_{IH} = \frac{V_{DD} + V_{T0,p} + k_R(2V_{out} + V_{T0,n})}{1 + k_R} \qquad (4)}$$

5. **Solve (3) and (4) together** for V_{IH} and $V_{out}\big|_{V_{IH}}$.



NMOS



PMOS

lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

WORKED EXAMPLE 23.1 · THE LECTURE METHOD, START TO FINISH

Take $V_{DD} = 5$ V, $V_{T0,n} = 1$ V, $V_{T0,p} = -1$ V, $k_n = 200$ and $k_p = 100$ μA/V², so $k_R = 2$.

V_{IL}. From ②, $V_{out} = \frac{1}{2}[3V_{IL} + 5 + 1 - 2] = 1.5V_{IL} + 2$, so $V_{out} - 5 = 1.5V_{IL} - 3$. Substitute into ① and write $x = V_{IL}$:

$$2(x-1)^2 = 2(x-4)(1.5x-3) - (1.5x-3)^2 = (1.5x-3)(0.5x-5)$$

$$2x^2 - 4x + 2 = 0.75x^2 - 9x + 15 \;\Rightarrow\; 1.25x^2 + 5x - 13 = 0$$

$$x = \frac{-5 + \sqrt{90}}{2.5} = \mathbf{1.795\ V}, \qquad V_{out} = 1.5x + 2 = 4.692\ \text{V}$$

V_{IH}. From ④, $V_{IH} = (6 + 4V_{out})/3$, so $V_{out} = 0.75V_{IH} - 1.5$. Substitute into ③:

$2\left[2(x-1)(0.75x-1.5) - (0.75x-1.5)^2\right] = (x-4)^2 \;\Rightarrow\; 0.875x^2 + 3.5x - 14.5 = 0 \;\Rightarrow\; x = \mathbf{2.536\ V}$

The margins are $NM_L = V_{IL} = 1.795$ V and $NM_H = 5 - 2.536 = 2.464$ V. The stronger NMOS pulls both V_{IL} and V_{IH} to the left.

SHORTCUT WORTH KNOWING (IT CHECKS YOUR QUADRATIC)

Eliminating V_{out} by hand gives V_{IL}, V_{IH} and V_{th} all in **one shape**:

$$V_x = \frac{V_{T0,n} + \rho\,(V_{DD} + V_{T0,p})}{1 + \rho}, \qquad \rho_{th} = \frac{1}{\sqrt{k_R}}, \quad \rho_{IL} = \frac{2\sqrt{k_R(k_R+3)} - k_R}{k_R(k_R+4)}, \quad \rho_{IH} = \frac{1 + 2\sqrt{1+3k_R}}{3k_R}$$

Here ρ measures how far you have moved from $V_{T0,n}$ toward $V_{DD} + V_{T0,p}$. With $k_R = 1$ you get $\rho = 1,\ 3/5,\ 5/3$, which gives the symmetric results of Lecture 24. For $k_R = 2$: $\rho_{IL} = 0.6180$ and $V_{IL} = (1 + 0.618 \times 4)/1.618 = 1.795$ V ✓. In an exam, show the lecture method. Use the shortcut to check your answer.

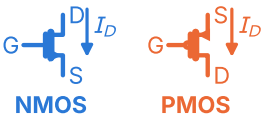
YOUR TURN · PART 1

1.1 Take $V_{DD} = 3.3$ V, $V_{T0,n} = 0.6$ V, $V_{T0,p} = -0.7$ V, $k_n = 200$ μA/V² and $k_p = 80$ μA/V². Find V_{OH}, V_{OL}, V_{IL}, V_{IH} and both noise margins. Inverter tutorial Q3, as listed in your Lec 18–22 guide

1.2 For the inverter of 1.1, name the region of each transistor at $V_{in} = 0.4$, 1.0, 2.0 and 3.0 V. You are told $V_{out} = 3.3$, 3.170, 0.052 and 0 V respectively. Check each answer against the boundary lines of §1.2.

1.3 Explain without equations why $V_{OH} = V_{DD}$ *exactly* in a CMOS inverter, while a resistive-load inverter has $V_{OL} > 0$.

1.4 Razavi §3.3.4, Eq. (3.48), Table 2.1 data. Bias an inverter at its trip point and it becomes an amplifier ("CS stage with active load"). Take $V_{DD} = 3$ V, $V_{THN} = 0.7$ V, $|V_{THP}| = 0.8$ V, $\mu_n C_{ox} = 134.3$ and $\mu_p C_{ox} = 38.4$ μA/V², $\lambda_n = 0.1$ and $\lambda_p = 0.2$ V^{-1}, and $(W/L)_n = 10/0.5$. (a) Choose $(W/L)_p$ so that $V_{th} = 1.5$ V. (b) Using $A_v = -(g_{m1} + g_{m2})(r_{O1} \parallel r_{O2})$, find the small-signal gain at V_{th}.



lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

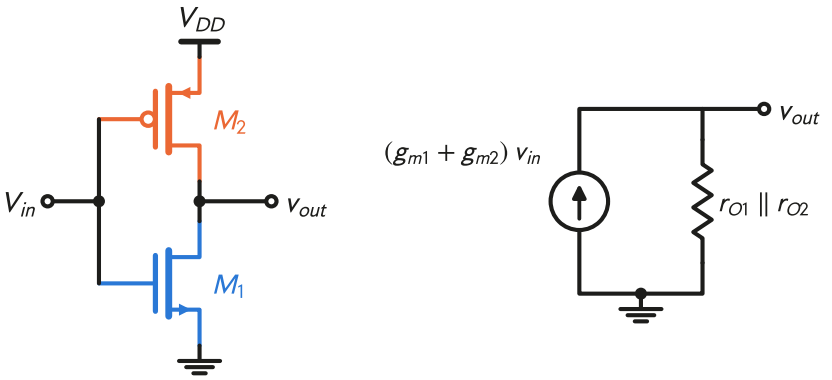


Fig. 1.4 For Q1.4: near V_{th} both devices are saturated and in parallel for small signals. Their transconductances add, and so do their output conductances (Razavi Fig. 3.20).

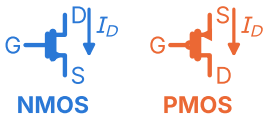
Solutions · Part 1

1.1 $V_{OH} = 3.3$ V and $V_{OL} = 0$ V. $k_R = 2.5$. V_{IL}: ② gives $V_{out} = \frac{1}{2}[3.5V_{IL} + 3.3 + 0.7 - 1.5] = 1.75V_{IL} + 1.25$. Substituting into ① gives a quadratic with root $V_{IL} = 1.065$ **V** ($V_{out} = 3.113$ V). V_{IH}: ④ gives $V_{out} = (3.5V_{IH} - 2.6 - 1.5)/5 = 0.7V_{IH} - 0.82$. Substituting into ③ gives $V_{IH} = 1.553$ **V** ($V_{out} = 0.267$ V). $NM_L = 1.065$ **V and** $NM_H = 1.747$ **V.** The shortcut check: $\rho_{IL} = 0.4266$ gives $V_{IL} = (0.6 + 0.4266 \times 2.6)/1.4266 = 1.065$ ✓.

1.2 At 0.4 V the NMOS is off and the PMOS is linear (point A). At 1.0 V: is $3.170 \geq 1.0 - 0.6$? Yes, so the NMOS is **saturated**. Is $3.170 \leq 1.0 + 0.7 = 1.7$? No, so the PMOS is **linear** (region B). At 2.0 V: is $0.052 \geq 1.4$? No, so the NMOS is linear. Is $0.052 \leq 2.7$? Yes, so the PMOS is saturated (region D). At 3.0 V: $3.0 > 3.3 - 0.7 = 2.6$, so the PMOS is off and the NMOS is linear (point E).

1.3 At $V_{in} = 0$ the NMOS is off, so no current can flow anywhere. A linear PMOS carrying zero current has zero V_{DS}, so $V_{out} = V_{DD}$. In the resistive inverter at $V_{in} = V_{OH}$, the NMOS is on and current flows through R_L. The output settles where the NMOS current equals $(V_{DD} - V_{out})/R_L$, which is above 0.

1.4 (a) From Lecture 24's inverse formula, $k_R = \left(\frac{3 - 0.8 - 1.5}{1.5 - 0.7}\right)^2 = 0.7656$. Then $k_n = 134.3\,\mu \times 20 = 2.686$ mA/V² and $k_p = k_n/k_R = 3.508$ mA/V², so $(W/L)_p = 3.508\,\text{m}/38.4\,\mu = 91.4 \approx 45.7/0.5$. (b) $I_D = \frac{k_n}{2}(0.8)^2 = 0.860$ mA, $g_{m1} = k_n(0.8) = 2.149$ mS and $g_{m2} = k_p(0.7) = 2.456$ mS. $r_{O1} = 1/(0.1\,I_D) = 11.63$ kΩ and $r_{O2} = 5.82$ kΩ, so their parallel combination is 3.88 kΩ. $A_v = -(4.605\text{ mS})(3.88\text{ kΩ}) \approx -17.9$. This finite gain is why the real VTC at C is steep but not vertical. With $\lambda = 0$, the gain would be infinite.



lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

PART 2 · LECTURE 24

Switching threshold, sizing and the symmetric inverter

One square root gives V_{th}. Run it backwards to size the transistors.

2.1 Deriving V_{th} (point C: both saturated)

1. **Equate the currents**, both saturated:

$$\frac{k_n}{2}(V_{GS,n} - V_{T0,n})^2 = \frac{k_p}{2}(V_{GS,p} - V_{T0,p})^2 \;\Rightarrow\; k_R(V_{in} - V_{T0,n})^2 = (V_{in} - V_{DD} - V_{T0,p})^2$$

2. **Put $V_{in} = V_{th}$ and take square roots:** $\sqrt{k_R}\,(V_{th} - V_{T0,n}) = \pm(V_{th} - V_{DD} - V_{T0,p})$.

   Which sign? The left side is positive (the NMOS is on). Inside the bracket, $V_{th} - V_{DD} - V_{T0,p} = V_{GS,p} - V_{T0,p} < 0$ (the PMOS is on). So take the **minus** sign: $\sqrt{k_R}(V_{th} - V_{T0,n}) = V_{DD} + V_{T0,p} - V_{th}$.

3. **Collect the V_{th} terms:** $V_{th}(1 + \sqrt{k_R}) = V_{DD} + V_{T0,p} + \sqrt{k_R}V_{T0,n}$. Divide top and bottom by $\sqrt{k_R}$:

$$\boxed{V_{th} = \frac{V_{T0,n} + \sqrt{1/k_R}\,(V_{DD} + V_{T0,p})}{1 + \sqrt{1/k_R}}}$$

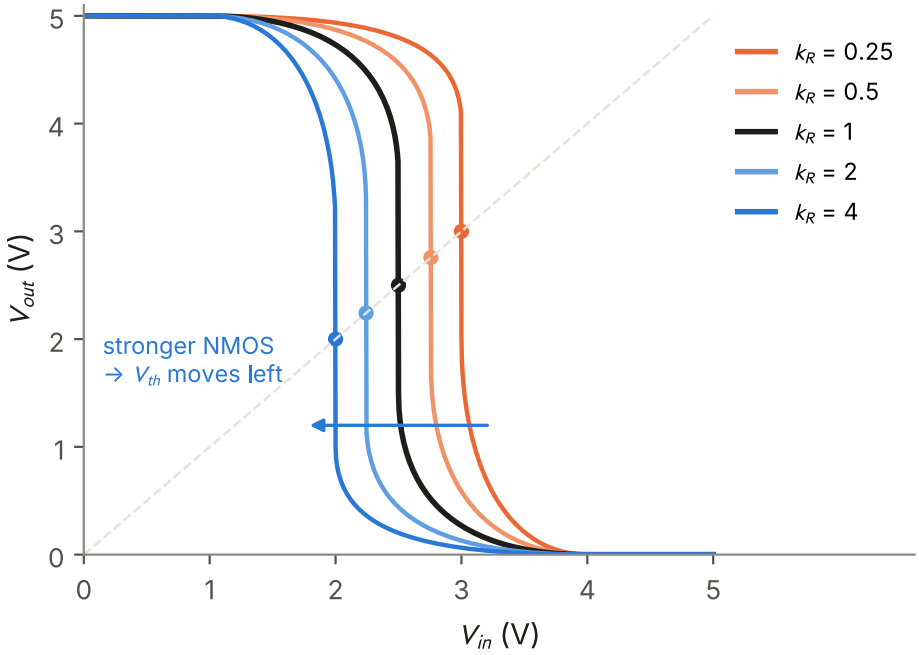
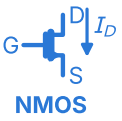


Fig. 2.1 VTCs for five values of k_R ($V_{DD} = 5$ V, $V_T = \pm 1$ V). The dots mark V_{th}. A stronger NMOS ($k_R > 1$) wins the tug-of-war earlier, so the curve shifts left. This matches the lecture sketch: $k_R > 1$ left, $k_R = 1$ middle, $k_R < 1$ right.

READ THE FORMULA AS A WEIGHTED AVERAGE

V_{th} always lies between $V_{T0,n}$ (where the NMOS starts to turn on) and $V_{DD} + V_{T0,p}$ (where the PMOS turns off). The weight on the PMOS end is $\sqrt{1/k_R}$. As $k_R \to \infty$, $V_{th} \to V_{T0,n}$. As $k_R \to 0$, $V_{th} \to V_{DD} + V_{T0,p}$.



lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

2.2 Design: run it backwards

Go back to step 2 and solve for k_R instead:

$$\sqrt{k_R} = \frac{V_{DD} + V_{T0,p} - V_{th}}{V_{th} - V_{T0,n}} \xrightarrow{V_{th} = V_{DD}/2} \boxed{k_R = \left(\frac{0.5V_{DD} + V_{T0,p}}{0.5V_{DD} - V_{T0,n}}\right)^2}$$

If $V_{T0,n} = |V_{T0,p}|$, the bracket is 1, so $k_R = 1$. Now expand k_R. With equal t_{ox}, $C_{ox} = \varepsilon_{ox}/t_{ox}$ cancels:

$$\frac{k_n}{k_p} = \frac{\mu_n C_{ox}(W/L)_n}{\mu_p C_{ox}(W/L)_p} = 1 \;\Rightarrow\; \frac{(W/L)_n}{(W/L)_p} = \frac{\mu_p}{\mu_n} = \frac{230}{580} \;\Rightarrow\; \boxed{(W/L)_p \approx 2.5\,(W/L)_n}$$

SYMMETRIC INVERTER

$V_{T0,n} = |V_{T0,p}|$ and $k_R = 1$ give $V_{th} = V_{DD}/2$, and the PMOS is about 2.5× wider to make up for its lower hole mobility.

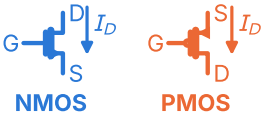
2.3 The symmetric inverter's V_{IL} and V_{IH}

Put $k_R = 1$ and $V_{T0,p} = -V_{T0,n} = -V_T$ into ②. The thresholds cancel:

$$V_{IL} = \frac{2V_{out} - V_T - V_{DD} + V_T}{2} = V_{out} - 0.5V_{DD} \quad \Rightarrow \quad V_{out} - V_{DD} = V_{IL} - 0.5V_{DD}$$

The lecture then writes "substitute into ①" and jumps straight to the answer. Here is the algebra in between. Write $x = V_{IL}$:

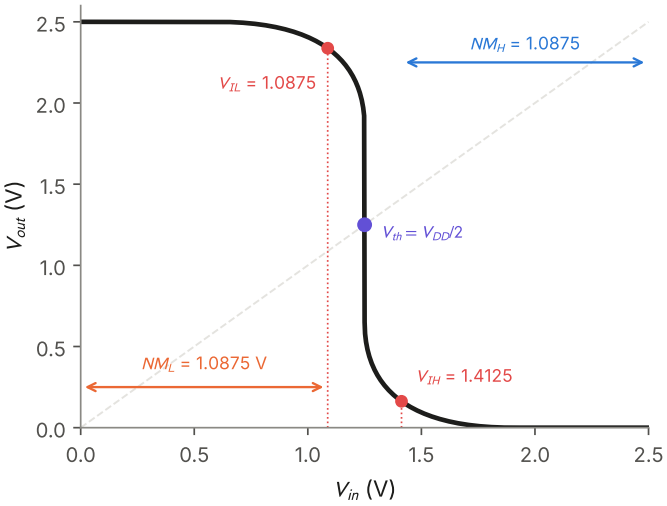
1. ① becomes $(x - V_T)^2 = 2(x - V_{DD} + V_T)(x - \frac{V_{DD}}{2}) - (x - \frac{V_{DD}}{2})^2$.
2. Expand both sides. The x^2 terms cancel:
$$x^2 - 2V_T x + V_T^2 = x^2 - 2V_{DD}x + 2V_T x + \tfrac{3}{4}V_{DD}^2 - V_T V_{DD}$$
3. Collect: $x(2V_{DD} - 4V_T) = \frac{3}{4}V_{DD}^2 - V_T V_{DD} - V_T^2 = \frac{1}{4}(3V_{DD} + 2V_T)(V_{DD} - 2V_T)$.
4. Cancel $(V_{DD} - 2V_T)$: $\boxed{V_{IL} = \frac{1}{8}(3V_{DD} + 2V_T)}$. The same steps on ③ and ④ give $V_{IH} = 0.5V_{DD} + V_{out}$ and $\boxed{V_{IH} = \frac{1}{8}(5V_{DD} - 2V_T)}$.



lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$



SYMMETRIC RESULTS

$$V_{IL} = \frac{3V_{DD} + 2V_T}{8}$$

$$V_{IH} = \frac{5V_{DD} - 2V_T}{8}$$

$$V_{IL} + V_{IH} = V_{DD}$$

$$NM_L = NM_H = V_{IL}$$

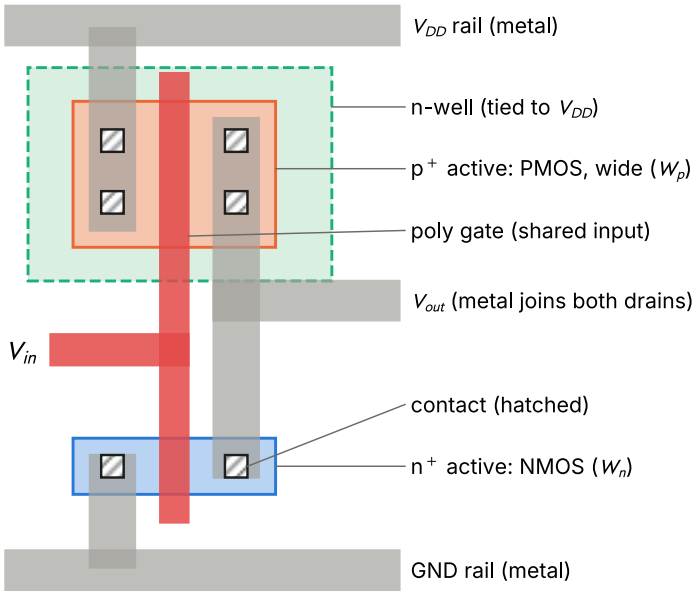
REMEMBER "3 + 5 = 8"

Low gets 3, high gets 5, both over 8. The $\pm 2V_T$ signs make the pair add to V_{DD}.

Fig. 2.2 A symmetric inverter ($V_{DD} = 2.5$ V, $V_T = 0.6$ V). The VTC is point-symmetric about $(V_{DD}/2, V_{DD}/2)$, so the two noise margins are equal. That is the reason to build it this way.

Why the margins are equal: $NM_L = V_{IL} - 0$, and $NM_H = V_{DD} - V_{IH} = V_{DD} - (V_{DD} - V_{IL}) = V_{IL}$.

2.4 Layout

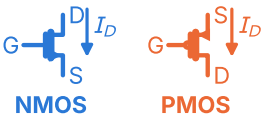


One vertical poly line is the common gate (V_{in}). Where it crosses the p⁺ region (inside the n-well) it forms the PMOS. Where it crosses the n⁺ region it forms the NMOS.

The sources contact the V_{DD} and GND rails. A metal strap joins the two drains to form V_{out}.

The p⁺ region is drawn about 2.5× wider. That is §2.2's sizing made visible.

Fig. 2.3 CMOS inverter layout, top view (after the lecture's sketch).



lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

YOUR TURN · PART 2

2.1 Inverter tutorial Q4 (2023-24) = 2024-25 Tutorial 5 Q2. Take $V_{DD} = 3.3$ V. NMOS: $V_{T0,n} = 0.6$ V, $\mu_n C_{ox} = 60\,\mu$ A/V², $(W/L)_n = 8$. PMOS: $V_{T0,p} = -0.7$ V, $\mu_p C_{ox} = 25\,\mu$A/V², $(W/L)_p = 12$. Find V_{th} and the noise margins.

2.2 Compre 2024-25 Q4(b). An inverter is sized for equal rise and fall delays: $V_{DD} = 2.5$ V, $V_{Tn} = -V_{Tp} = 0.6$ V, $k'_n = 3k'_p = 120\,\mu$A/V². Find the noise margins. *(Part 3 explains why equal delays mean $k_n = k_p$.)*

2.3 Take $V_{DD} = 3.3$ V, $V_{T0,n} = 0.6$ V, $V_{T0,p} = -0.7$ V and $\mu_n/\mu_p = 580/230$, with equal t_{ox}. Find $(W/L)_p/(W/L)_n$ for $V_{th} = V_{DD}/2$.

2.4 A symmetric 5 V inverter ($V_T = 1$ V) is redesigned with $k_R = 4$. Find the new V_{th}. Do the margins become equal, or does one grow at the other's expense?

2.5 Prove $V_{IL} + V_{IH} = V_{DD}$ for the symmetric inverter *without* the closed forms. Use the 180° symmetry of the VTC.

Solutions · Part 2

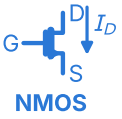
2.1 $k_n = 480$ and $k_p = 300\ \mu$A/V², so $k_R = 1.6$ and $\sqrt{1/k_R} = 0.7906$. $V_{th} = \dfrac{0.6 + 0.7906 \times 2.6}{1.7906} = 1.483$ **V**. For V_{IL}, ② gives $V_{out} = \frac{1}{2}(2.6V_{IL} + 3.04)$. Substituting into ① gives $V_{IL} = 1.199$ **V** $= NM_L$. For V_{IH}: $V_{IH} = 1.696$ **V, so** $NM_H = 1.604$ **V**. The printed key gives 1.48, 1.16, 1.69 and 1.61 V. A brute-force sweep of the VTC confirms 1.199 V for V_{IL}, so the key's 1.16 is slightly off. Both were accepted.

2.2 Equal delays need equal strengths, so $(W/L)_p = 3(W/L)_n$, giving $k_R = 1$ with matched thresholds: the inverter is symmetric. $V_{IL} = (7.5 + 1.2)/8 = 1.0875$ V and $V_{IH} = (12.5 - 1.2)/8 = 1.4125$ V. $NM_L = NM_H = 1.0875$ **V** and $V_{th} = 1.25$ V.

2.3 $k_R = \left(\dfrac{1.65 - 0.7}{1.65 - 0.6}\right)^2 = (0.9048)^2 = 0.8186$. Then $\dfrac{(W/L)_p}{(W/L)_n} = \dfrac{\mu_n}{\mu_p\, k_R} = \dfrac{2.522}{0.8186} =$ **3.08**. The larger $|V_{T0,p}|$ weakens the PMOS, so it must be made even wider than 2.5×.

2.4 $V_{th} = \dfrac{1 + 0.5 \times 4}{1.5} = 2.0$ **V**. The shortcut gives $V_{IL} = 1.512$ V and $V_{IH} = 2.219$ V, so $NM_L = 1.51$ **V and** $NM_H = 2.78$ **V**. They are no longer equal: NM_H grows while NM_L shrinks from 1.75 V. The symmetric design maximises the *worse* of the two margins.

2.5 In the symmetric case the PMOS is the NMOS with every voltage measured from the other rail. So if (V_{in}, V_{out}) lies on the VTC, so does $(V_{DD} - V_{in}, V_{DD} - V_{out})$. That map sends a slope of −1 to a slope of −1, so it carries the point V_{IL} to the point V_{IH}. Therefore $V_{IH} = V_{DD} - V_{IL}$.



NMOS



PMOS

lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

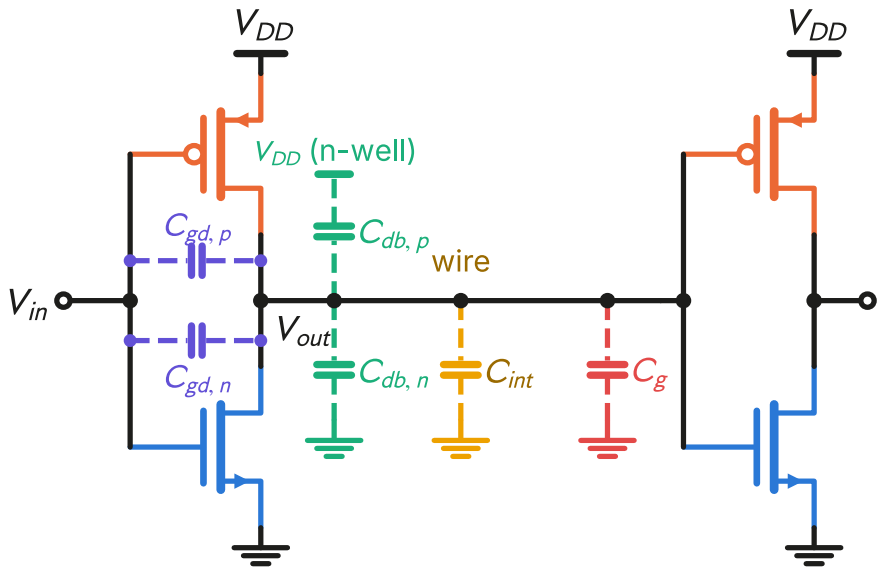
$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

PART 3 · LECTURE 25

Switching characteristics and delay

The output is a capacitor. The transistors are the currents that charge and discharge it.

3.1 Where the load capacitance comes from



$C_{gs,p}$, $C_{gs,n}$ hang on the input node (or are shorted): not on V_{out}

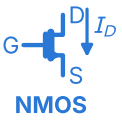
next stage: $C_g = C_{ox}(WL)_n + C_{ox}(WL)_p$

Fig. 3.1 The lecture's picture: the parasitics of an inverter driving a second inverter. Every capacitor with one plate on V_{out} counts toward the load. C_{gs} sits between the gate and a fixed rail, so it loads the *input*, not the output.

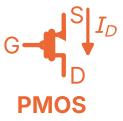
LUMPED LOAD

$$C_{load} = C_{gd,p} + C_{gd,n} + C_{db,p} + C_{db,n} + C_{int} + C_g$$

C_{gd} is gate–drain overlap. C_{db} is the drain–bulk junction. C_{int} is the wire. C_g is the next stage's gate capacitance: $C_g \approx C_{ox}\,[(WL)_n + (WL)_p]$.



NMOS



PMOS

lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

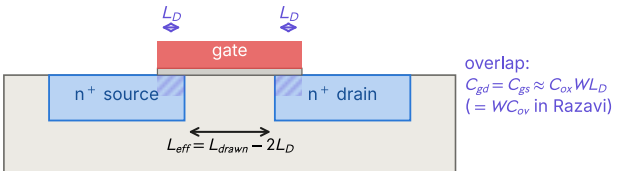


Fig. 3.2 The source and drain diffuse a distance L_D under the gate. That overlap is C_{gd} and C_{gs} (the lecture's L_D sketch).

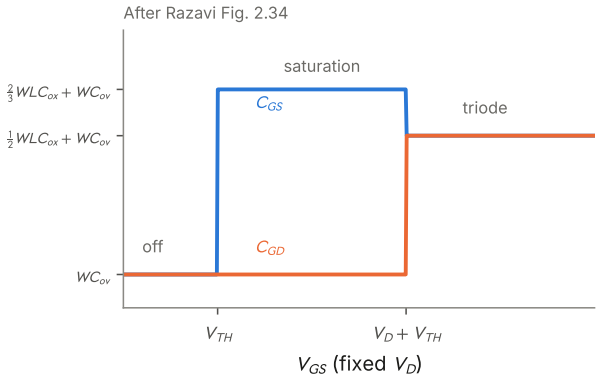


Fig. 3.3 Razavi's bias dependence. In saturation, the channel connects to the source, so C_{GD} is overlap only.

From Razavi §2.4.2: $C_{gd} = WC_{ov}$ $(= C_{ox}WL_D)$. The junction capacitance is $C_{db} = C_j \cdot (\text{drain area}) + C_{jsw} \cdot (\text{drain perimeter})$. A folded layout (Razavi Ex. 2.10) halves the drain area and so shrinks C_{db}.

3.2 The lumped model

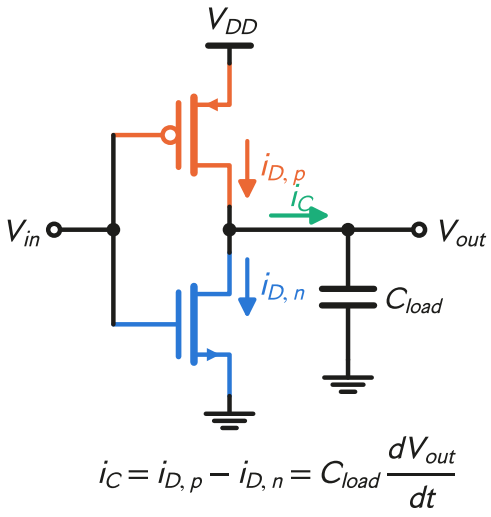


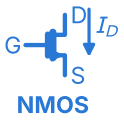
Fig. 3.4 Everything on the output node lumped into C_{load}.

All the physics collapses to one node equation (KCL at V_{out}):

$$i_C = i_{D,p} - i_{D,n} = C_{load}\frac{dV_{out}}{dt}$$

Falling output (V_{in} steps high): the PMOS turns off, so $i_C = -i_{D,n}$ and the NMOS discharges C_{load}.

Rising output (V_{in} steps low): the NMOS turns off, so the PMOS charges C_{load}.





lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

3.3 Definitions

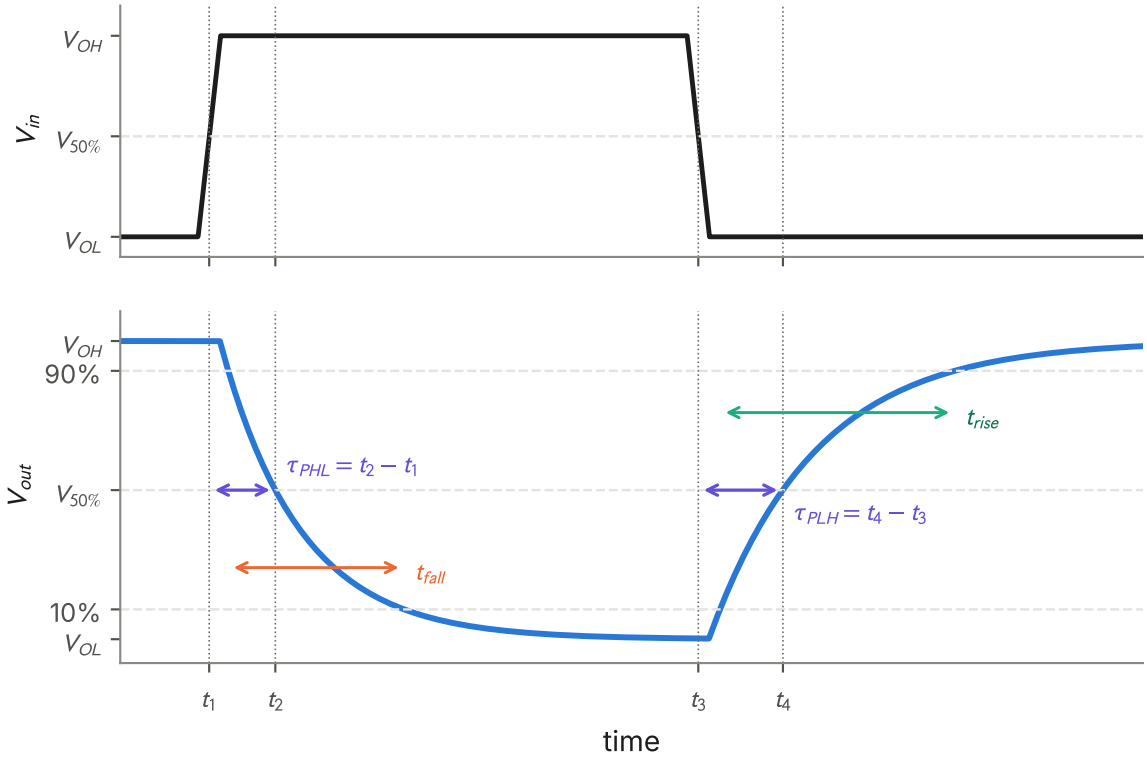


Fig. 3.5 Delays are measured between the 50 % points of input and output. Rise and fall times are measured on the output alone, between 10 % and 90 %.

QUANTITY	DEFINITION
$V_{50\%}$	$V_{OL} + \frac{1}{2}(V_{OH} - V_{OL}) = \dfrac{V_{OH} + V_{OL}}{2}$
τ_{PHL} (output high → low)	$t_2 - t_1$
τ_{PLH} (output low → high)	$t_4 - t_3$
τ_P (average)	$\dfrac{\tau_{PHL} + \tau_{PLH}}{2}$
rise time / fall time	$t_{90\%} - t_{10\%}$ (output L → H) / $t_{10\%} - t_{90\%}$ (output H → L)

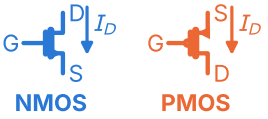
3.4 Method 1: average current

Pretend the capacitor current is constant at its average value. Then $i = C\,dV/dt$ becomes $\tau = C\,\Delta V / I_{avg}$:

$$\tau_{PHL} = \frac{C_{load}\,\Delta V_{HL}}{I_{avg,HL}}, \qquad \Delta V_{HL} = V_{OH} - V_{50\%} = \frac{V_{OH} - V_{OL}}{2}$$

$$I_{avg,HL} = \tfrac{1}{2}\left[i_C(V_{in} = V_{OH},\, V_{out} = V_{OH}) + i_C(V_{in} = V_{OH},\, V_{out} = V_{50\%})\right]$$

Likewise $\tau_{PLH} = C_{load}\Delta V_{LH}/I_{avg,LH}$, with $\Delta V_{LH} = V_{50\%} - V_{OL}$, and the current averaged at $V_{out} = V_{OL}$ and $V_{out} = V_{50\%}$ (with $V_{in} = V_{OL}$).



lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

PICK THE RIGHT REGION FOR EACH ENDPOINT

Consider the falling output. At its start ($V_{out} = V_{OH}$), $V_{DS,n} = V_{OH} \geq V_{OH} - V_{T0,n}$, so the NMOS is **saturated**. At $V_{50\%}$, normally $V_{50\%} < V_{OH} - V_{T0,n}$, so it is **linear**. Using the saturation formula at both ends is the most common mistake.

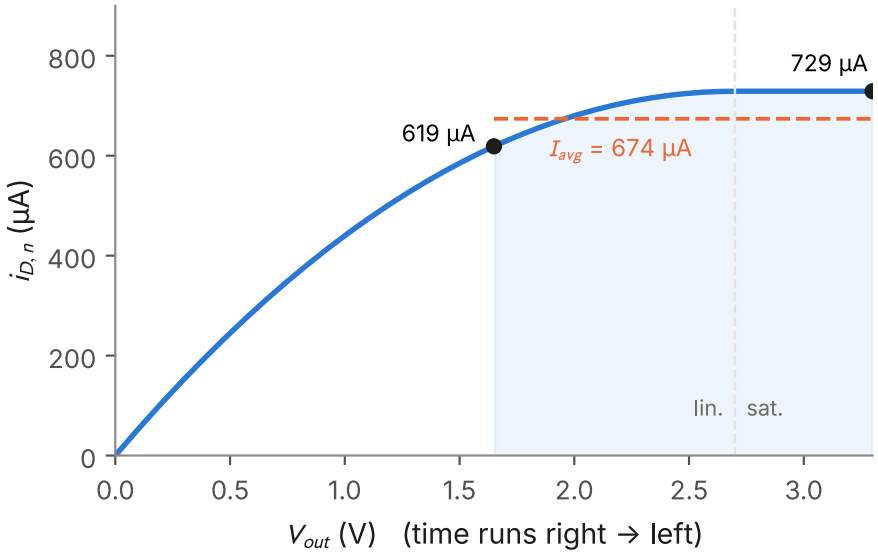


Fig. 3.6 The NMOS current during the fall. The method replaces the real curve (shaded) by the mean of its two end values.

3.5 Method 2: state equations (exact integration)

Step the input from V_{OL} to V_{OH} at t_1. As V_{out} falls, the NMOS passes through **two regions**, so the integral splits in two:

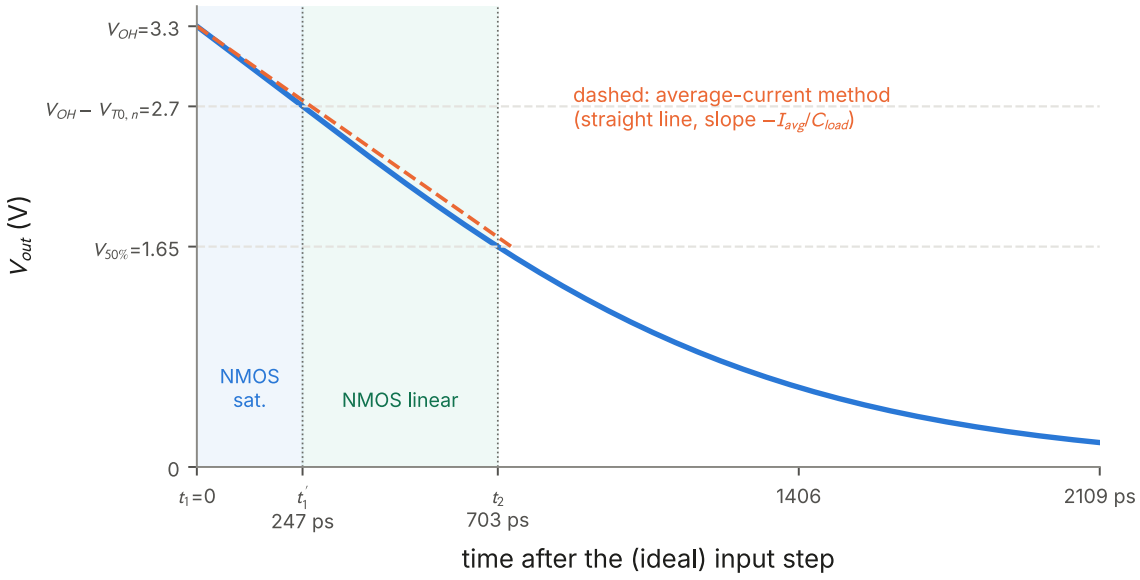
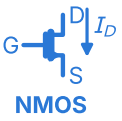


Fig. 3.7 The exact $V_{out}(t)$ (Worked example 25.2's numbers). From t_1 to t_1' the NMOS is saturated and the discharge is a straight line (constant current). From t_1' to t_2 it is linear and the discharge curves.



lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

Segment 1: saturated, $V_{OH} \to V_{OH} - V_{T0,n}$ (in the lecture)

The NMOS stays saturated while $V_{out} \geq V_{in} - V_{T0,n} = V_{OH} - V_{T0,n}$. Its current is the constant $\frac{k_n}{2}(V_{OH} - V_{T0,n})^2$:

$$dt = -\frac{C_{load}\,dV_{out}}{i_{D,n}} \;\Rightarrow\; \int_{t_1}^{t_1'} dt = \frac{-C_{load}}{\frac{k_n}{2}(V_{OH} - V_{T0,n})^2}\int_{V_{OH}}^{V_{OH}-V_{T0,n}} dV_{out} \;\Rightarrow\; \boxed{t_1' - t_1 = \frac{2C_{load}V_{T0,n}}{k_n(V_{OH} - V_{T0,n})^2}}$$

Segment 2: linear, $V_{OH} - V_{T0,n} \to V_{50\%}$ (completing the lecture)

Now $i_{D,n} = \frac{k_n}{2}\left[2(V_{OH} - V_{T0,n})V_{out} - V_{out}^2\right]$. Write $a = 2(V_{OH} - V_{T0,n})$, so the bracket is $V_{out}(a - V_{out})$:

$$t_2 - t_1' = -\frac{2C_{load}}{k_n}\int_{V_{OH}-V_{T0,n}}^{V_{50\%}} \frac{dV_{out}}{V_{out}(a - V_{out})} = \frac{2C_{load}}{k_n\,a}\left[\ln\frac{V}{a - V}\right]_{V_{50\%}}^{V_{OH}-V_{T0,n}}$$

At the upper limit $V = a/2$, so $\ln\frac{V}{a-V} = \ln 1 = 0$. That leaves

$$t_2 - t_1' = \frac{C_{load}}{k_n(V_{OH} - V_{T0,n})}\ln\left(\frac{2(V_{OH} - V_{T0,n})}{V_{50\%}} - 1\right)$$

MEMORISE: EXACT STEP-INPUT DELAYS

$$\tau_{PHL} = \frac{C_{load}}{k_n(V_{OH} - V_{T0,n})}\left[\frac{2V_{T0,n}}{V_{OH} - V_{T0,n}} + \ln\left(\frac{4(V_{OH} - V_{T0,n})}{V_{OH} + V_{OL}} - 1\right)\right]$$

$$\tau_{PLH} = \frac{C_{load}}{k_p(V_{OH} - V_{OL} - |V_{T0,p}|)}\left[\frac{2|V_{T0,p}|}{V_{OH} - V_{OL} - |V_{T0,p}|} + \ln\left(\frac{4(V_{OH} - V_{OL} - |V_{T0,p}|)}{V_{OH} - V_{OL}} - 1\right)\right]$$

For CMOS ($V_{OH} = V_{DD}$, $V_{OL} = 0$), both have the form $\frac{C}{k\,(V_{DD} - |V_T|)}\left[\frac{2|V_T|}{V_{DD}-|V_T|} + \ln\left(\frac{4(V_{DD}-|V_T|)}{V_{DD}} - 1\right)\right]$. The PLH formula is the PHL one with $n \to p$ and voltages measured down from V_{DD}.

WORKED EXAMPLE 25.2

Take $V_{DD} = 3.3$ V, $C_{load} = 0.3$ pF, $k_n = 200$ and $k_p = 80$ μA/V², $V_{T0,n} = 0.6$ V, $V_{T0,p} = -0.7$ V.

Average current, HL. At $V_{out} = 3.3$ (sat.): $100\,\mu(2.7)^2 = 729$ μA. At $V_{out} = 1.65$ (lin.): $100\,\mu[2(2.7)(1.65) - 1.65^2] = 619$ μA. The average is 674 μA, so $\tau_{PHL} = 0.3\text{p} \times 1.65/674\,\mu = \mathbf{0.734}$ ns.

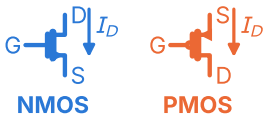
Exact, HL. $\frac{0.3\text{p}}{200\mu \times 2.7} = 0.556$ ns, times $\left[\frac{1.2}{2.7} + \ln\left(\frac{10.8}{3.3} - 1\right)\right] = 0.444 + 0.821$, gives $\tau_{PHL} = 0.247 + 0.456 = \mathbf{0.703}$ ns.

Exact, LH. $\frac{0.3\text{p}}{80\mu \times 2.6}\left[\frac{1.4}{2.6} + \ln\left(\frac{10.4}{3.3} - 1\right)\right] = 1.442 \times (0.538 + 0.766)$, so $\tau_{PLH} = \mathbf{1.88}$ ns (the average-current method gives 1.96 ns). Then $\tau_P = 1.29$ ns.

The average-current method overestimates by about 4 %. That is good enough for most questions, but use whichever method the question names.

3.6 What the formulas tell a designer

Both delays have the form $\tau \propto \frac{C_{load}}{k\,(V_{DD} - |V_T|)}$. Razavi's one-line estimate (§17.1) is $T_d \approx CV_{DD}/I$. To go faster:



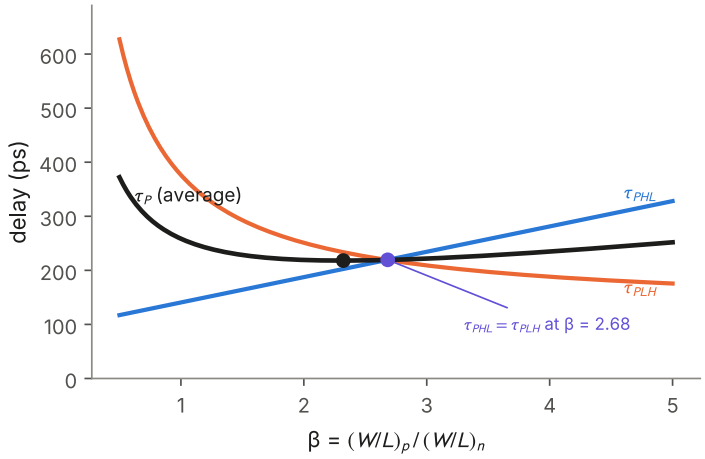
lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

- **Reduce C_{load}:** shorter wires, smaller fan-out, folded drains.
- **Increase k:** make W/L larger. But wider devices add their own C_{db} and load the previous stage more, so the gain is limited.
- **Increase V_{DD}:** faster, but power $\propto CV_{DD}^2 f$ rises.

Equal rise and fall need $k_p = k_n$ (with matched thresholds). That is the same $(W/L)_p \approx 2.5(W/L)_n$ as Lecture 24's symmetric inverter, which is why Compre Q4 said "sized for equal delays" and meant "symmetric".



The plot drives an identical next stage, so a wider PMOS also adds load capacitance. τ_{PLH} falls as the PMOS widens, but τ_{PHL} rises (more capacitance, same NMOS).

The delays are equal at $\beta \approx 2.7$. The *average* is smallest a little earlier, at $\beta \approx 2.3$ (black dot). Equal edges and minimum average delay are slightly different goals.

Fig. 3.8 Delay against PMOS/NMOS sizing, with self-loading (illustrative values).

YOUR TURN · PART 3

3.1 Take $C_{ox} = 3.8$ fF/µm², $L_D = 0.08$ µm, $L = 0.5$ µm, $W_n = 1$ µm and $W_p = 2.5$ µm. Also $C_{db,n} = 2$ fF, $C_{db,p} = 4$ fF and $C_{int} = 5$ fF. The inverter drives an identical inverter. Find C_{load}. Which term dominates?

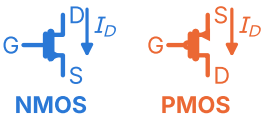
3.2 Derive segment 2 of §3.5 yourself, starting from $C\,dV_{out}/dt = -i_{D,n}$ in the linear region. Check that you reach the ln term.

3.3 In Worked example 25.2, what $(W/L)_p$ multiplier makes $\tau_{PLH} = \tau_{PHL}$? Assume C_{load} does not change.

3.4 Razavi Problem 2.11(b), with numbers added. An NMOS ($W/L = 50/0.5$, $\mu_n C_{ox} = 134.3\,\mu$A/V², $V_{TH} = 0.7$ V, $\lambda = \gamma = 0$) has its source grounded and its drain on node X. $C_1 = 1$ pF runs from X to ground and starts at 1 V. At $t = 0$ the gate steps from 0 to 3 V. Sketch $V_X(t)$ and find when $V_X = 0.5$ V.

3.5 Razavi Problem 2.11(a), with numbers added. Same device, but now V_{DD} is on the drain and C_1 (initially 1 V) is on the source. The gate steps from 0 to 3 V. Sketch $V_X(t)$, give its final value, and find when $V_X = 2.0$ V.

3.6 Razavi §15.2. A ring of 5 inverters, each with $\tau_P = 50$ ps, oscillates. What is its frequency?



lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ sat : $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ sat : $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

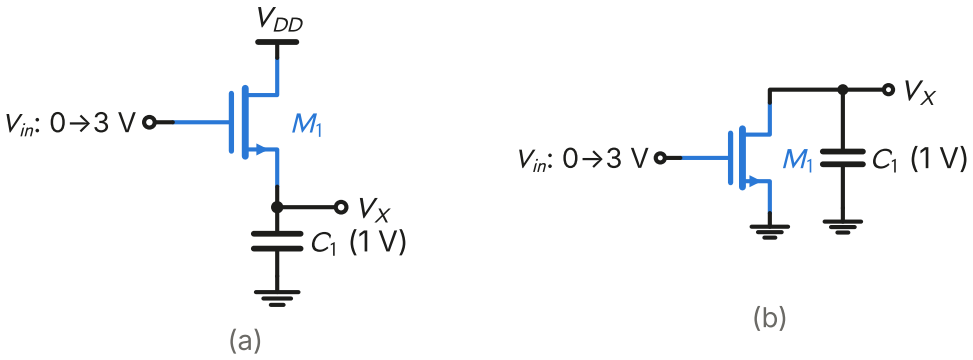


Fig. 3.9 Razavi Fig. 2.53(a) and (b), redrawn for Q3.4 and Q3.5.

Solutions · Part 3

3.1 $C_{gd,n} = 3.8 \times 1 \times 0.08 = 0.30$ fF and $C_{gd,p} = 3.8 \times 2.5 \times 0.08 = 0.76$ fF. $C_g = 3.8 \times (1 + 2.5) \times 0.5 = 6.65$ fF. $C_{load} = 0.30 + 0.76 + 2 + 4 + 5 + 6.65 = 18.7$ **fF**. The next stage's gate dominates, closely followed by the wire.

Some texts double C_{gd} for the Miller effect: both ends of it swing in opposite directions. The lecture counts it once.

3.2 See §3.5. The keys are the partial fractions $\frac{1}{V(a-V)} = \frac{1}{a}\left(\frac{1}{V} + \frac{1}{a-V}\right)$, and noticing the upper limit is $a/2$, which kills one log.

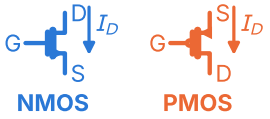
3.3 Equate the exact formulas: $\tau_{PLH} \propto 1/k_p$, so scale k_p by $1.882/0.703 =$ **2.68**, i.e. $(W/L)_p$ about 2.7× its current value. (That is larger than $k_n/k_p = 2.5$ because $|V_{T0,p}| > V_{T0,n}$.)

3.4 $V_{GS} = 3$ V and $V_{DS} \leq 1 < 2.3$ V, so the device is **linear the whole time**. V_X falls toward 0 like a slightly bent exponential (Fig. 3.10b). This is segment 2 alone:

$$t = \frac{C}{k_n V_{ov}} \ln \frac{V_0\,(2V_{ov} - V)}{V\,(2V_{ov} - V_0)}, \quad k_n = 13.43\ \text{mA/V}^2,\ V_{ov} = 2.3\ \text{V}$$

$\frac{C}{k_n V_{ov}} = 32.4$ ps and $\ln \frac{1 \times 4.1}{0.5 \times 3.6} = 0.823$, so $t \approx 26.6$ **ps**.

3.5 $V_{DS} = 3 - V_X \geq V_{GS} - V_{TH} = 3 - V_X - 0.7$ always holds, so the device is **saturated** throughout. It is a source follower charging C_1: $C\,dV_X/dt = \frac{k_n}{2}(2.3 - V_X)^2$. V_X rises toward $V_{in} - V_{TH} = 2.3$ **V** but never quite reaches it, because the current dies as $V_{GS} \to V_{TH}$. Integrating, $\frac{1}{2.3 - V}\Big|_1^V = \frac{k_n t}{2C}$, so $t = \frac{2C}{k_n}\left(\frac{1}{0.3} - \frac{1}{1.3}\right) =$ **0.38 ns**. (With $\gamma > 0$, the final value is lower, because V_{TH} rises with V_{SB}.)



lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

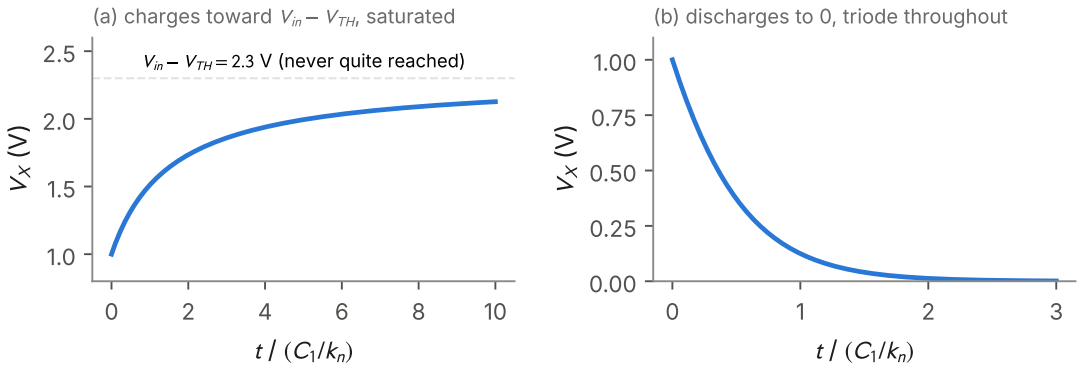
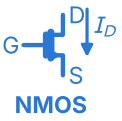


Fig. 3.10 The answers to Q3.5 (a) and Q3.4 (b), with time normalised to C_1/k_n.

3.6 An edge must travel round the loop twice for one period, through N inverters each time: $T = 2N\tau_P = 500$ ps, so $f = 2$ **GHz**. (The number of stages must be odd, or the ring latches instead of oscillating.)





lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

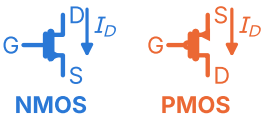
$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

PART 4

Formula sheet and revision

Everything from Lectures 23–25 that you should be able to write from memory

RESULT	FORMULA	CONDITIONS
Boundary lines	$V_{out} = V_{in} - V_{T0,n}$; $V_{out} = V_{in} - V_{T0,p}$	NMOS sat. above ; PMOS sat. below
V_{OH}, V_{OL}	V_{DD}, 0	always (CMOS)
V_{IL} (with ①)	$\frac{2V_{out} + V_{T0,p} - V_{DD} + k_R V_{T0,n}}{1 + k_R}$	n sat., p lin., slope −1
V_{IH} (with ③)	$\frac{V_{DD} + V_{T0,p} + k_R(2V_{out} + V_{T0,n})}{1 + k_R}$	n lin., p sat., slope −1
V_{th}	$\frac{V_{T0,n} + \sqrt{1/k_R}(V_{DD} + V_{T0,p})}{1 + \sqrt{1/k_R}}$	both sat.
k_R for a target V_{th}	$\left(\frac{V_{DD} + V_{T0,p} - V_{th}}{V_{th} - V_{T0,n}}\right)^2$	
Symmetric sizing	$(W/L)_p = \frac{\mu_n}{\mu_p}(W/L)_n \approx 2.5\,(W/L)_n$	$V_{T0,n} = \lvert V_{T0,p}\rvert$, $k_R = 1$
Symmetric V_{IL}, V_{IH}	$\frac{1}{8}(3V_{DD} + 2V_T)$, $\frac{1}{8}(5V_{DD} - 2V_T)$; $NM_L = NM_H = V_{IL}$	$k_R = 1$
Noise margins	$NM_L = V_{IL} - V_{OL}$, $NM_H = V_{OH} - V_{IH}$	
Load	$C_{gd,p} + C_{gd,n} + C_{db,p} + C_{db,n} + C_{int} + C_g$	$C_{gd} = C_{ox} W L_D$
$V_{50\%}$ and delays	$\frac{1}{2}(V_{OH} + V_{OL})$; $\tau_P = \frac{1}{2}(\tau_{PHL} + \tau_{PLH})$	50 % to 50 %
Average-current	$\tau = C_{load}\Delta V / I_{avg}$, $\Delta V = \frac{1}{2}(V_{OH} - V_{OL})$	average of the 2 endpoints
Saturated segment	$t_1' - t_1 = \frac{2C_{load}V_{T0,n}}{k_n(V_{OH} - V_{T0,n})^2}$	$V_{OH} \to V_{OH} - V_{T0,n}$
Exact τ_{PHL}	$\frac{C_{load}}{k_n(V_{OH} - V_{T0,n})}\left[\frac{2V_{T0,n}}{V_{OH} - V_{T0,n}} + \ln\left(\frac{4(V_{OH} - V_{T0,n})}{V_{OH} + V_{OL}} - 1\right)\right]$	step input
Exact τ_{PLH}	same, with $n \to p$ and $V_{OH} - V_{T0,n} \to V_{OH} - V_{OL} - \lvert V_{T0,p}\rvert$, $V_{OH} + V_{OL} \to V_{OH} - V_{OL}$	step input



lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

Memory tricks

REGIONS A–E

"Off-Lin, Sat-Lin, Sat-Sat, Lin-Sat, Lin-Off", NMOS first. The middle three spell a palindrome around C.

V_IL VS V_IH FORMULAS

Both have $1 + k_R$ below and $2V_{out}$ on top. In V_{IL}, k_R multiplies only $V_{T0,n}$. In V_{IH}, it multiplies $(2V_{out} + V_{T0,n})$, because the NMOS now sets the output.

V_TH AS A TUG-OF-WAR

It sits between $V_{T0,n}$ and $V_{DD} + V_{T0,p}$. The stronger side pulls V_{th} toward its own threshold.

DELAY

"C over kV, times [sat. bit + log bit]." The sat. bit is $2V_T/(V_{DD} - V_T)$. The log bit is $\ln(4 \cdot \text{overdrive}/V_{DD} - 1)$.

Flashcards (cover the right half)

Point B: which device is saturated?

The NMOS (the PMOS is linear).

Why is V_{OL} exactly 0?

The PMOS is off, so there is no current, so the linear NMOS has $V_{DS} = 0$.

Which sign after the square root for V_{th}?

Minus: $V_{GS,p} - V_{T0,p} < 0$.

$k_R > 1$ moves the VTC...

Left (lower V_{th}).

$(W/L)_p$ for a symmetric inverter?

$\mu_n/\mu_p \approx 580/230 \approx 2.5$ times $(W/L)_n$.

$V_{IL} + V_{IH}$ (symmetric)?

V_{DD}.

Is C_{gs} part of C_{load}?

No. It loads the input node.

NMOS region at $V_{out} = V_{50\%}$ during the fall?

Linear (if $V_{50\%} < V_{OH} - V_{T0,n}$).

$t_1' - t_1$?

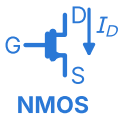
$2C_{load}V_{T0,n}/[k_n(V_{OH} - V_{T0,n})^2]$

Ring of N inverters: period?

$2N\tau_P$ (N odd).

Exam playbook

1. **Any CMOS VTC question:** compute $k_n = \mu_n C_{ox}(W/L)_n$, k_p and k_R first. Write the four terminal voltages, then V_{th} from the boxed formula.



NMOS



PMOS

lin : $V_{DS} < V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}[2(V_{GS} - V_{T0,n})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \geq V_{GS} - V_{T0,n}$ $I_D = \frac{k_n}{2}(V_{GS} - V_{T0,n})^2$

lin : $V_{DS} > V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}[2(V_{GS} - V_{T0,p})V_{DS} - V_{DS}^2]$ **sat :** $V_{DS} \leq V_{GS} - V_{T0,p}$ $I_D = \frac{k_p}{2}(V_{GS} - V_{T0,p})^2$

$k = \mu C_{ox}\frac{W}{L}$, $k_R = \frac{k_n}{k_p}$, $V_{T0,p} < 0$ inverter: $V_{GS,n} = V_{in}$, $V_{DS,n} = V_{out}$, $V_{GS,p} = V_{in} - V_{DD}$, $V_{DS,p} = V_{out} - V_{DD}$

2. V_{IL} / V_{IH}**:** write ① or ③, differentiate, set the slope to −1, solve for V_{out} in terms of V_{IL} (or V_{IH}), substitute, and solve the quadratic. Check the answer with the ρ shortcut. If the inverter is symmetric, use the $3/8$ and $5/8$ forms directly.
3. **Design for** V_{th}**:** $k_R = \left(\frac{V_{DD}+V_{T0,p}-V_{th}}{V_{th}-V_{T0,n}}\right)^2$, then convert to W/L using the mobility ratio. "Equal rise/fall" means symmetric.
4. **Delay:** write C_{load}, decide each transistor's region at both endpoints, then use average current or the exact formula, whichever the question asks for. Always state $V_{50\%}$.